
ATML Assignment 1

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Abstract

Deep neural networks frequently fail under deployment conditions because real-world data does not always follow the same feature distribution as the training data. This performance drop can occur due to a model's inductive biases and differences in the underlying data distributions. This report evaluates the robustness and adaptability of deep neural networks through multiple experiments: exploring inherent inductive biases via controlled interventions (like color and cue-conflict), target-aware domain adaptation (DAN, DANN, CDAN), target-free domain generalization (DAN-DG, SAM), and open-set recognition (GCSC, PROSER). Our findings show that contrastive text-supervision and heavy augmentations make models more robust and improve far-unknown rejection. Furthermore, explicitly aligning domains can lead to negative transfer when the source and target class distributions are inherently different, whereas using the average of multiple source domains yields more robust generalization.

Code for all experiments is available at: <https://github.com/aitasaamjaf123/Inductive-Biases-DA-DG-UDA>.

1 Introduction

Deep neural networks achieve great performance on closed-world, independently and identically distributed (i.i.d.) vision datasets, yet they frequently fail when deployed in real-world environments. Standard empirical risk minimization (ERM) models often learn the path of least resistance, relying on spurious correlations, such as local texture and background shortcuts, instead of robust global structures. While existing methodologies attempt to solve these problems separately, there remains a critical need to systematically evaluate how varying architectures (CNNs vs. Vision Transformers), training techniques, and alignment strategies behave across domain shifts. This report empirically investigates model robustness using four different approaches: shape-texture/color/positional bias analysis inherent in a model, target-aware domain adaptation, target-free domain generalization, and open-set recognition.

Research questions.

- **RQ1:** How do different model architectures (like CNNs vs. ViTs) and training methods affect a model's reliance on shape versus texture?
- **RQ2:** Does aligning features across domains actually improve target performance, or does it cause negative transfer when the source and target class distributions don't match?
- **RQ3:** How well can different novelty detection methods and training techniques separate known classes from near and far unknowns?

Contributions.

- **Robustness & Bias Analysis:** Demonstrated that text-supervised models (CLIP) are forced to align representations more strictly under spatial disturbances compared to ERM-trained CNNs, which tend to memorize local visual shortcuts.

- **Domain Adaptation vs. Generalization:** Revealed that target free domain generalization (DAN-DG) can, counter-intuitively, outperform target-aware adaptation (DAN, CDAN), primarily because target-aware alignment can fail severely when source and target label distributions are heavily mismatched.
- **Open-Set Recognition Benchmarking:** Empirically established that heavy random augmentations (GCSC) can significantly improve the detection of far-unknown classes by penalizing shortcut learning, whereas explicitly modeling the unknown space with dummy classifiers (PROSER) can overlap with and degrade known class boundaries.

2 Experimental Setup / Methods

2.1 Datasets and Domains

The dataset chosen for Task 1 is STL-10 due to lower computational cost. Using the PACS dataset in Task 2, Sketch is used as the target domain, whereas art painting, cartoon, and photo are chosen as the source domains. Labeled data from the source domains and unlabeled target data are used for training. In task 3, no target data is used, and the model is trained entirely on the source domains for better generalization. In this experiment, CIFAR-10 is used as the known dataset and CIFAR-100 as the unknown dataset, with near and far unknowns pre-defined from CIFAR-100.

2.2 Model Architectures

For task 1, a linear classifier is trained on top of RESNET-50, ViT-B/16 and OpenClip BiT-B-32, along with evaluations using CLIP zero-shot. Task 2 evaluates domain adaptability using a ResNet-18 architecture across four distinct domains. Task 3 explores domain generalization using the same ResNet-18 across four distinct domains. Task 4 explores open-set recognition using ResNet-18.

2.3 Methods and Interventions Compared

Task 1. Palette transfer is chosen as the additional color intervention as it maps one image’s color onto the other without transferring any major features, instead of a fixed change of color in every image, and hence it is easier to compare model outputs of different images. Initially, the cue-conflict class pairs were chosen to map animals to material objects, but it led to a low coverage due the model being confused between two very different classes. Hence, the following pairs were chosen to ensure cue-conflict pairs are relatively more closer to each other in the feature space. This was done by pairing animals with animals and material objects together: Dog-horse, deer-horse, cat-dog, bird-monkey, car-truck. (This pairing increased the coverage from 50-60% to 70-80%). The style strength was chosen to be 1 for complete style transfer. Otherwise, our results would have been more shape-biased, as texture would not have been completely transferred.

Task 2. We explored the effect of three different adaptation techniques: DAN (uses MMD to align source and target representations) [Long et al., 2015], DANN (uses a gradient reversal layer to push the representations together; same underlying principle as DAN, but using a different technique) [Ganin et al., 2016], and CDAN (tries to make sure $P(Y|X)$, the class conditional, is also aligned in addition to the alignment of features only) [Long et al., 2018] across the same ResNet-18 model. The features (right after the encoder backbone) were normalized for both DANN and CDAN to ensure training stability. Because at the beginning of training, random weight initialization can cause the loss to be unstable, normalizing the features ensured both DANN and CDAN showed stable training.

Task 3. Two new methodologies were used: DAN-DG (the same DAN implemented in Task 2, but this time the MMD is computed between different source domains and then averaged for each pair, using the same kernels) [Long et al., 2015] and SAM (this follows a different principle; it tries to find a point on the loss curve where the curve is flat, so that a small perturbation does not increase the loss by a huge factor) [Foret et al., 2021].

Task 4. This task explores multiple techniques (GCSC [Vaze et al., 2022], PROSER [Zhou et al., 2021], and RPL [Chen et al., 2020]) to ensure a model is able to recognize an image that does not belong to the image distributions or domains it was trained on.

2.4 Hyperparameters

Global & Data: Seed 6304, 4 workers, 224×224 resolution

Feature Extraction: Batch size 256

Linear Probing: AdamW optimizer, $\text{lr}=1 \times 10^{-3}$, weight decay= 1×10^{-4} , batch size 128, max epochs 50, early stopping patience 5

Evaluation: Train/Val split 0.2 (4000 train, 1000 val), balanced test subset of 50 images/class (500 total), zero-shot batch size 256

Interventions: Palette transfer target pairs 500, cue conflict candidates 40/direction (400 total), AdaIN $\alpha = 1.0$, translation displacements 0, 8, 16, 32 px, patch shuffle grid 4 (16 patches)

3 Results for Tasks 1–4

3.1 Task 1: Robustness and Shape–Texture Bias in Vision Backbones

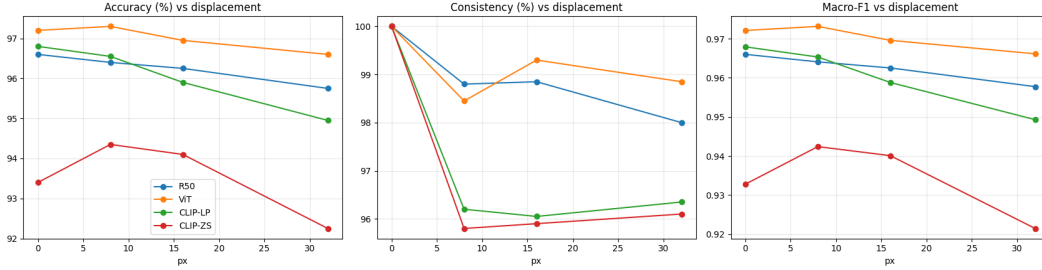


Figure 1: Displacement plots measuring representation changes.

As seen from Table 1, the grayscale and palette transfer interventions cause minimal accuracy drops across all models (ranging from -0.4% to -4.4%), which indicates that none of the models rely heavily on color for classification. Accuracy drops minimally under the grayscale transformation for ResNet-50 (-1.6%), ViT-B/16 (-2.8%), OpenCLIP (-4.4%), and CLIP zero-shot (-3.6%). Palette transfer yields similarly negligible accuracy reductions for ResNet-50 (-2.0%), ViT-B/16 (-0.8%), OpenCLIP (-2.2%), and CLIP zero-shot (-0.4%). However, across every model except ResNet, performance is better with palette transfer relative to the grayscale intervention, as indicated by the consistency and accuracy values. In other words, the models, on average, appear more sensitive to color removal than to color change under this setting.

The cue-conflict experiments, on the other hand, reveal a heavy reliance on shape across all architectures [Geirhos et al., 2019], as seen in Table 2. Figure 2 provides selected examples of these generated pairs. All models exhibit a shape bias above 70%, with a coverage of around 70–80%. However, ResNet has a lower shape bias and coverage relative to the other models. The reason is that ResNet relies more heavily on local features and textures due to its convolutional architecture. In contrast, for example, the early layers in ViT focus on global structure (due to the attention mechanism). The shape bias conclusions are robust due to the high coverage percentages, with ResNet at 72.578% and all other models showing a coverage above 80%.

When spatial structure is altered via translation, all models on average show gradual degradation in their accuracy (approximately -3%) and prediction consistency (approximately -4%) with an increase in displacement (0–32px). This indicates that none of the models are perfectly invariant to translation, as illustrated in Figure 1.

Patch-shuffling, on the other hand, drastically separates the models based on their reliance on global structure. ResNet-50 and ViT-B/16 show a relatively smaller drop in performance compared to the CLIP models, as indicated in Table 1. Even though all models rely on global structure to some extent, text-supervised models (OpenCLIP and CLIP zero-shot) are forced to learn images against descriptive text captions due to contrastive loss. Whereas, ResNet-50, for example, due to its reliance on cross-entropy loss, always tries to take the path of least resistance, and hence these models can memorize local visual patterns like an ear, etc.

Table 1: Task 1 primary metrics across models and interventions.

Experiment	Metric	ResNet-50	ViT-B/16	OpenCLIP	CLIP (zero-shot)
Clean	acc (%)	96.6	97.2	96.8	93.4
	macro F1	0.966	0.972	0.968	0.933
	mean conf	0.923	0.965	0.303	0.926
	consist. (%)	100.0	100.0	100.0	100.0
Grayscale	acc (%)	95.0 (-1.6)	94.4 (-2.8)	92.4 (-4.4)	89.8 (-3.6)
	macro F1	0.950 (-.016)	0.944 (-.028)	0.924 (-.044)	0.898 (-.035)
	mean conf	0.921 (-.003)	0.954 (-.011)	0.277 (-.026)	0.914 (-.013)
	consist. (%)	95.2	96.6	93.4	92.4
Palette transfer	acc (%)	94.6 (-2.0)	96.4 (-0.8)	94.6 (-2.2)	93.0 (-0.4)
	macro F1	0.946 (-.020)	0.964 (-.008)	0.946 (-.022)	0.928 (-.004)
	mean conf	0.917 (-.006)	0.956 (-.010)	0.287 (-.016)	0.919 (-.007)
	consist. (%)	96.0	98.0	97.2	96.0
Patch shuffle	acc (%)	89.4 (-7.2)	91.2 (-6.0)	82.6 (-14.2)	80.2 (-13.2)
	macro F1	0.895 (-.071)	0.913 (-.059)	0.824 (-.144)	0.799 (-.134)
	mean conf	0.883 (-.041)	0.898 (-.068)	0.197 (-.106)	0.754 (-.172)
	consist. (%)	91.2	93.4	82.6	81.2

Table 2: Shape, texture, and other decision counts alongside shape bias and coverage.

Model	Shape	Texture	Other	Total	Shape Bias (%)	Coverage (%)
ResNet-50 (linear probe)	114	39	57	210	74.510	72.857
ViT-B/16 (linear probe)	150	21	39	210	87.719	81.429
OpenCLIP (linear probe)	152	18	40	210	89.412	80.952
CLIP (zero-shot)	147	24	39	210	85.965	81.429

For ResNet and ViT, prediction consistency does not strictly agree with the changes in feature representations. Specifically, the cue-conflict and patch-shuffling experiment results do not align closely with representation stability. This implies that for both ResNet and ViT, the decision boundary is wide enough to maintain high consistency and coverage. However, since the loss does not directly penalize features being far apart, these models can get away with feature representations that still lead to high prediction accuracy. However, for the CLIP models, since the loss includes cosine similarity, they are forced to align their feature representations, as seen in Table 3 and Figure 3.

OpenCLIP demonstrates better results than CLIP zero-shot across all interventions because the linear classifier head for OpenCLIP was explicitly trained for this task; hence, this training bias is a reason for the better performance of OpenCLIP with a linear probe.

Table 3: Representation stability across all interventions.

Condition	ResNet-50	ViT-B/16	OpenCLIP ViT-B/32
grayscale	0.798	0.732	0.885
palette	0.872	0.872	0.926
cue_conflict	0.412	0.409	0.770
translate_8	0.967	0.945	0.978
translate_16	0.951	0.976	0.964
translate_32	0.931	0.944	0.967
patch_shuffle	0.695	0.633	0.751



Figure 2: Selected examples of generated cue-conflict image pairs mapping animals and material objects.

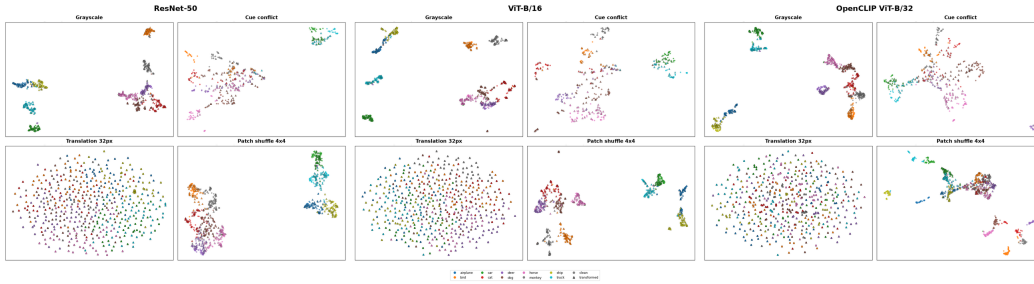


Figure 3: UMAP projection comparing clean and transformed features in the latent space.

3.2 Task 2: Domain Adaptation (DAN, DANN, CDAN)

The training curves in Figure 4 indicate that all four mechanisms achieved stable training. As previously mentioned in the experimental setup, stable training for DANN and CDAN was achieved using normalization.

Before concluding anything, it is important to first establish how different and far apart the source and target distributions are. A class distribution comparison done in Appendix A.1, Figure 8 indicates that the class distribution in the source is distributed almost evenly across all seven classes (and a bit skewed towards house and person). On the other hand, the target majorly comprises the animal classes and the guitar class, whereas the house and person classes occupy a smaller fraction. This distribution will help in our analysis later on.

Table 4: Comparison of domain adaptation methods on source and target domains.

Method	Source Acc	Source F1	Target Acc	Target F1	Target Acc Δ	Domain Separability
Source-only	0.940	0.938	0.730	0.744	0.000	0.996
DAN	0.933	0.930	0.626	0.547	-0.104	0.902
DANN	0.932	0.929	0.425	0.449	-0.305	1.000
CDAN	0.935	0.935	0.070	0.076	-0.660	0.999

Table 5: Per-source validation domain performance (Accuracy and Macro-F1) before pooling.

Method	Photo		Art Painting		Cartoon	
	Acc	F1	Acc	F1	Acc	F1
Source-only	0.970	0.964	0.907	0.903	0.942	0.947
DAN	0.970	0.964	0.895	0.893	0.934	0.932
DANN	0.961	0.953	0.905	0.902	0.930	0.932
CDAN	0.952	0.946	0.898	0.901	0.955	0.959

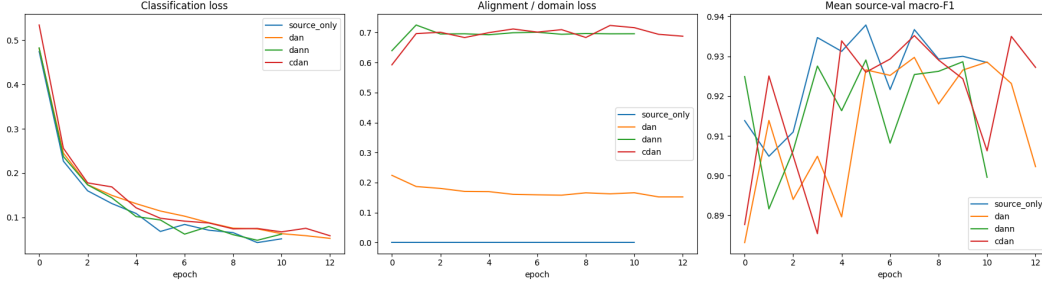


Figure 4: Classification, domain-loss/alignment, and mean source-val macro f1 training curves assessing adaptation method convergence.

The Source-only ERM has a domain separability of 0.996, which means that source and target distributions can be easily separated by a discriminator, and it can also be seen in the UMAP plots in the Appendix. On the target domain, the accuracy of our source-only model drops by roughly 20% due to two main reasons. Firstly, due to the high domain separability of the source and target. And secondly, due to a mismatch in class-wise distributions. Since we did not enforce a class-balanced target set, our model trained on source images, even if it had learned domain-invariant features, would have performed badly on the target because our class distribution in the target is skewed towards animals. If a certain proportion of our feature space is occupied by a certain class, and there are fewer samples of that class in the target distribution, our model will ultimately perform worse. Hence, it makes sense to conclude that house and person are most likely the classes which are causing the highest proportion of failures on the target.

Across all three methods, only DAN achieved some amount of domain mixing (0.902, still not that significant), as seen in Table 4. Even for DAN, the target accuracy decreased by roughly 10%. CDAN and DANN, on the other hand, failed to mix the domains and led to negative transfer, as seen by their accuracies in Table 4. Class-level comparisons provided in Figure 5 and Table 6 provide empirical evidence for this conclusion.

The controlled alignment study was performed for DAN due to its ability to align source and target domains relative to DANN. Prior to evaluation, we hypothesize that increasing the alignment strength (λ_{MMD}) will successfully decrease domain separability by forcing the source and target feature distributions closer together. However, because the source and target domains possess

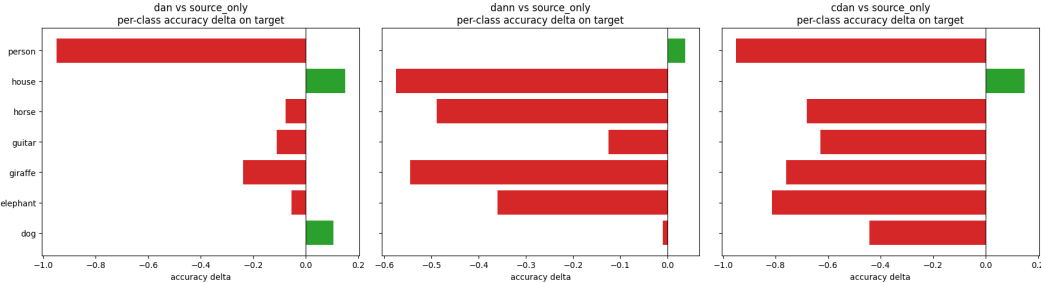


Figure 5: Per-class target accuracy changes highlighting specific positive and negative transfer effects.

Table 6: Most degraded classes vs. source-only and top failure cases (confusions).

Method	Class	Δ Acc	Top Confusion
DAN	person	-0.950	house (160 times)
	giraffe	-0.239	house (120 times)
	guitar	-0.110	house (83 times)
DANN	house	-0.575	person (54 times)
	giraffe	-0.546	person (413 times)
	horse	-0.489	person (391 times)
CDAN	person	-0.950	house (160 times)
	elephant	-0.812	house (740 times)
	giraffe	-0.760	house (748 times)

Table 7: Controlled alignment-strength study for DAN evaluating varying λ_{MMD} .

λ_{MMD}	Source Acc	Target Acc	Target F1	Domain Separability
0.1	0.927	0.653	0.600	0.988
1.0	0.933	0.626	0.547	0.902
10.0	0.879	0.603	0.608	0.901

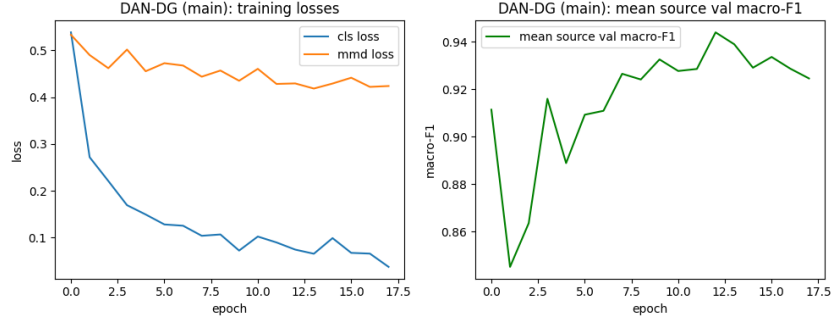
fundamentally different class distributions (source dominated by house/person; target dominated by animals), this forced alignment is expected to induce negative transfer. Consequently, higher alignment pressure should progressively degrade both source performance and target recognition as distinct class boundaries are incorrectly mixed. As seen in Table 7, sweeping lambda from 0.1 to 1.0 causes some target and source mixing, and this mixing, in turn, leads to a decreased accuracy. A lambda of 10.0 yields no further benefit to domain mixing and decreases the accuracy further. Hence, in our particular experiment, we can say that a decrease in accuracy is expected if we only align the distributions, because the class distributions are different for the source and target.

However, this should then imply that CDAN should show a better performance, but it turns out that it in fact shows a very high negative transfer. The reason for this is that since the class distributions in the source and target are very different as explained in the Appendix, CDAN forces target images to align with the majority classes (house and person) in the source. Since these same two classes are present in the least abundance in the target, the accuracy for CDAN is lower than both DAN and DANN, simply because it incorporates the class prior distribution as well, which acts as an add-on in decreasing the accuracy, and this effect is also visible in Figure 5, the per-class accuracy change. Furthermore, for CDAN, we did not implement entropy-based weighting of samples. Hence, another factor is that at the start of our training, the outputs of our classifier would be random, and the discriminator will still consider them valid. Because we are not weighting the “false” outputs of our classifier, our model can perform very badly, as indicated.

3.3 Task 3: Domain Generalization (DAN-DG, SAM)

Before any comparison with Task 2, it is important to note that even though the baseline ERM model of Task 3 was trained in the same settings as Task 2, its accuracy is actually lower. Task 2’s ERM accuracy was around 0.73, whereas Task 3’s source-only ERM accuracy dropped to 0.55. Hence, any comparison between Task 2 and 3 should consider this factor. Ideally, Task 3 should have used the same trained model from Task 2, but despite the fixed seed and same train/val split, the exact trajectory and early stopping can differ, which is evident in the change of accuracy.

From Table 8, it is evident that mean-source and worst-source validation accuracies are not a good predictor of target performance. DAN-DG achieves a target accuracy of 69.00%, whereas SAM achieves a target accuracy of 65.46%, which is opposite to the trend of mean and worst source accuracy across both of them. Aggregate source/target metrics are also not a good representative of better performance across every class. For example, from the class-wise accuracy of each model in Table 12, it can be seen that DAN-DG performs severely worse than SAM on dogs, but its mean accuracy is skewed due to large improvements in other classes. Prior to evaluation, we hypothesize



(a) DAN-DG: classification loss, MMD penalty, and mean source macro-F1 vs. epoch.



(b) SAM: classification loss and mean source macro-F1 vs. epoch (no explicit alignment penalty).

Figure 6: Training curves for the main DAN-DG ($\lambda_{DG} = 1$) and SAM ($\rho = 0.05$) runs. DAN-DG’s MMD loss plateaus around 0.42–0.48 while classification loss keeps falling, suggesting alignment saturates well before classification converges.

that increasing the domain generalization penalty (λ_{DG}) will decrease source-domain separability by forcing the representations of Photo, Art, and Cartoon to align. While moderate alignment is expected to improve generalization to the unseen Sketch domain, excessive alignment ($\lambda_{DG} = 10.0$) will likely over-constrain the network, strip away class-discriminative information, and severely degrade both source and target accuracy. The controlled study on λ_{DG} , as indicated in Table 11, shows how DAN-DG affects domain separability and sketch accuracy. At $\lambda = 0.1$, it does not mix the domains well (83.72% separability), yet it achieves the highest sketch accuracy (70.53%). At $\lambda = 1.0$, it achieves a much lower separability of 67.77%, but this actually corresponds to a decreased target accuracy of 53.35%. Furthermore, it is important to note that at $\lambda = 10$, it worsens both the separability and accuracy (source and target both). Hence, in this specific sweep, heavy domain invariance did not strictly improve sketch recognition. Class-level alignment is still not guaranteed. The model still performs worse than the baseline on dogs and horses at its suitable lambda value because it relies on the source domains without any incorporation of target class distributions in any way.

Table 8: ERM, DAN-DG, and SAM compared on each source-validation domain, mean-source, worst-source, and Sketch accuracy/macro-F1, with the change in Sketch accuracy relative to ERM.

Method	Photo	Art	Cartoon	Mean Acc	Worst Acc	Mean F1	Worst F1	Sketch Acc	Sketch F1	Δ Sketch Acc
ERM	96.41	90.00	91.26	92.56	90.00	92.31	89.81	55.08	49.32	—
DAN-DG	96.71	91.95	94.67	94.44	91.95	94.39	91.93	69.00	68.94	+13.92
SAM	98.20	92.68	95.10	95.33	92.68	95.40	92.70	65.46	65.83	+10.38

As evident in Table 9, SAM has the minimum Δ_{sharp} out of all three. Even though this does not directly indicate a globally flatter loss landscape, it is suitable to conclude that for the specific ϵ setting, SAM has a higher local stability than the baseline and DAN-DG. However, this doesn’t directly correspond to a higher sketch performance than DAN-DG, and it doesn’t align the source domains with each other much relative to the ERM model, which was expected. DAN-DG’s sharpness is also less than the baseline. This can mean that both sharpness and domain alignment can contribute to improving sketch performance. If we could make a model which flattens the loss curve and brings

the domains together without causing a class-shift, we would likely achieve better results than the individual models.

The results from Table 10 appear counter-intuitive to what one might expect. Task 2’s DAN, which had access to unlabeled sketch data, performs worse than Task 3’s target-free DAN-DG. Even though the ERM source accuracy for Task 3 is lower than Task 2, Task 3’s DAN-DG still has a higher overall sketch accuracy. Relative to its baseline, Task 3 outperforms Task 2’s DAN by a huge margin (+13.92 vs. -10.40).

One explanation for this behavior is that when training DAN with unlabeled target data, the uneven class distribution in the source and target forces target classes to be intermixed with source classes, such that target classes fall into a “wrong” region. However, this should not be a dominant effect, as we saw that Task 2 DAN did not show a very low domain separability (roughly 0.92). Hence, another explanation is that in the case of DAN-DG, we are computing MMD as the average of three different MMDs across three different domains. This increase in the number of domains for evaluating MMD is a likely reason for the better generalization of DAN-DG, as it forces the model to find a more universal representation across varied visual styles.

Table 9: Source-domain separability (logistic-regression accuracy predicting Photo/Art/Cartoon, chance = 33.3%) and the common sharpness proxy Δ_{sharp} (cross-entropy increase under a fixed $\rho = 0.05$ perturbation) for all three models.

Method	Source sep. (%)	Δ_{sharp}
ERM	89.37	0.3180
DAN-DG	69.10	0.1989
SAM	87.71	0.0905

Table 10: Target-aware DAN (Task 2, sees unlabeled Sketch during training) vs. target-free DAN-DG (Task 3, never accesses Sketch), both relative to their respective baselines.

Comparison	Baseline Acc	Method Acc	Δ
Task 2: Source-only $\rightarrow$ DAN	73.00	62.60	-10.40
Task 3: ERM $\rightarrow$ DAN-DG	55.08	69.00	+13.92

Table 11: Controlled study: DAN-DG with $\lambda_{\text{DG}} \in \{0.1, 1, 10\}$. All other settings fixed. Source metrics are mean/worst across Photo, Art, and Cartoon.

λ_{DG}	Mean Acc	Worst Acc	Separability (%)	Sketch Acc	Sketch F1
0.1	93.82	92.68	83.72	70.53	70.25
1.0	93.84	91.22	67.77	53.35	57.16
10.0	15.53	9.28	68.11	18.22	4.42

Table 12: Per-class Sketch accuracy (%) for ERM, DAN-DG, and SAM, with changes relative to ERM.

Class	ERM	DAN-DG	Δ DAN-DG	SAM	Δ SAM
dog	33.55	28.76	-4.79	51.68	+18.13
elephant	87.70	98.11	+10.41	97.57	+9.87
giraffe	89.24	73.04	-16.20	54.45	-34.79
guitar	49.51	85.53	+36.02	94.90	+45.39
horse	28.92	67.77	+38.85	44.36	+15.44
house	35.00	67.50	+32.50	75.00	+40.00
person	11.88	53.75	+41.87	26.25	+14.37

3.4 Task 4: Open-Set Recognition (OSR)

As evident from Table 13, semantic similarity majorly dominates the rejection pattern (e.g., Mahalanobis: 0.799 near vs. 0.915 far AUROC). Semantically similar classes are inherently harder to differentiate compared to far classes.

Our Vanilla model shows two kinds of failures: one on near classes and one on far classes. On near classes, a failure is expected, as seen in Table 14. A cat is predicted for a wolf, and an automobile or truck is predicted for a pickup truck, which makes sense intuitively as they are semantically related classes and have a similar texture, shape, and overall distribution of pixels. Confident predictions on far classes are, however, more concerning, as they stem from the model not being able to learn useful features and relying on shortcuts instead. A chair is confused for an airplane, and a bottle for a bird; neither of these pairs has any close semantic relationship, yet the model confuses them, which implies that the model is relying on spurious features for its predictions. The models are evaluated

Table 13: Open-set recognition performance of post-hoc novelty scores (MSP, MLS, Energy, and Mahalanobis) evaluated on the frozen Vanilla ResNet-18 baseline.

Score Method	AUROC (Near)	AUROC (Far)	AUROC (All)	FPR@95TPR (Near)	FPR@95TPR (Far)	FPR@95TPR (All)
MSP	0.813	0.891	0.852	0.699	0.583	0.641
MLS	0.795	0.883	0.839	0.673	0.496	0.584
Energy	0.795	0.884	0.840	0.664	0.481	0.573
Mahalanobis	0.799	0.915	0.857	0.706	0.495	0.601

Table 14: Informative unknown failure cases misclassified as known classes under the Vanilla MLS validation-calibrated threshold ($\tau = -5.77$).

Difficulty	True Unknown Class	Predicted Known Class	Score	Threshold (τ)	Status (False Positive)
Near	wolf	cat	-7.51	-5.77	Accepted (High Confidence)
Near	pickup_truck	automobile	-7.26	-5.77	Accepted (High Confidence)
Near	pickup_truck	truck	-7.36	-5.77	Accepted (High Confidence)
Far	chair	airplane	-8.24	-5.77	Accepted (High Confidence)
Far	clock	ship	-5.93	-5.77	Accepted (High Confidence)
Far	bottle	bird	-8.05	-5.77	Accepted (High Confidence)

using multiple novelty scores: MSP, MLS, Energy, and Mahalanobis. MSP measures how confident our model is in its top guess (normalized) [Hendrycks and Gimpel, 2017], MLS retains absolute logit magnitude, Energy uses all logits [Liu et al., 2020], and Mahalanobis measures the distance from known-class feature clusters [Lee et al., 2018]. Looking at Table 13, it can be seen that AUROC and FPR disagree with each other. For AUROC, Mahalanobis performs best (0.915 for far and 0.799 for near), whereas for FPR, Energy acts as a better measure.

GCSC’s random augmentations helped improve both closed-set accuracy and open-set recognition. Known-class accuracy (CSA) went up slightly from 94.61% to 95.33%. Near-unknown detection improved too, but only slightly. Far-unknown detection improved significantly: AUROC rose from 0.883 to 0.915, and the rejection rate for far unknowns jumped from 50.4% to 61.5%. The reason for this is that heavy augmentation (cropping, changing color, etc.) forces the model not to rely on shortcuts and instead pushes the model to learn actual causal features. This helps distinguish far unknowns better. Even though this augmentation improves near unknowns, the improvement is only by a small factor; because near unknowns are closely related to actual classes, it remains very difficult for a model to classify animals with similar textures and shapes as different classes. Hence, augmentation mostly helps with improving far unknown predictions.

Intuitively, one should expect PROSER to improve near-end accuracy while decreasing or keeping far-end accuracy and source accuracy the same. But that is not what the results indicate. As seen in Table 15, PROSER performs worse on the knowns, the near unknowns, and the far unknowns compared to both Vanilla and GCSC. A possible explanation for this result is that since PROSER creates a “mix” of our known classes, many of our near-unknown class samples do not necessarily lie at the exact overlap of two different classes. For example, a fox may not necessarily lie close to the region between a cat and a dog. A stronger reason is that there are 8 near-unknown categories (and 8 far-unknown), but only 5 dummy classifiers; hence, the rejection classes might be overlapping excessively even if the features align. The higher T_{95} values indicate that PROSER did shift the decision boundaries, but this shifting did not ultimately help in classification.

Table 15: Closed-set accuracy (CSA) and open-set recognition metrics comparing Vanilla, GCSC, and PROSER using MLS and placeholder-based scores.

Method	Score Variant	CSA	AUROC _{near}	AUROC _{far}	AUROC _{all}	T_{95} Acc	Near Rej	Far Rej	FPR _{near}	FPR _{far}
Vanilla	MLS	0.9461	0.795	0.883	0.839	0.949	0.328	0.504	0.673	0.496
GCSC	MLS	0.9533	0.818	0.915	0.866	0.945	0.346	0.615	0.654	0.385
PROSER	MLS	0.9442	0.786	0.868	0.827	0.954	0.294	0.420	0.706	0.580
PROSER	Placeholder	0.9442	0.774	0.884	0.829	0.952	0.290	0.471	0.710	0.529

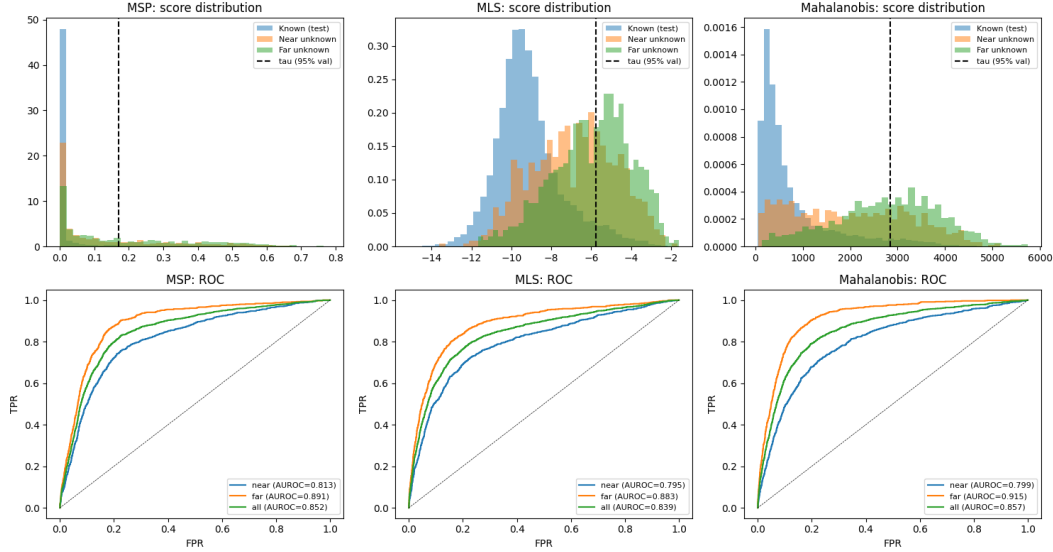


Figure 7: Score distributions and ROC curves for MSP, MLS, and Mahalanobis detectors under the frozen Vanilla model, separating known test samples from near and far unknown CIFAR-100 classes.

4 Cross-task Discussion

4.1 Visual Cues and Distribution Shift

It is observed that almost all of the evaluated models are shape-biased, with ResNet exhibiting a stronger texture and local-structure bias relative to the other architectures. Color interventions caused only a minuscule drop in performance. Tasks 2 and 3 introduced a severe domain shift, as the Sketch domain lacks both color and realistic texture. Because ResNet showed a dependence on texture in Task 1, we observe the baseline ERM dropping significantly in accuracy (73.0% in Task 2 and 55.08% in Task 3). This qualitative vulnerability is most pronounced for ResNet, as the other models were more shape-biased, as seen in Task 1.

4.2 Invariance, Discriminability, and Target Access

Across Tasks 2 and 3, making domains less distinguishable is achieved at the risk of negative transfer and diminished target performance. In Task 2, for example, CDAN forced target representations closer together despite the target images being skewed towards animals, while the source images were skewed towards “house” and “person” classes. This forced invariance caused massive negative transfer, dropping target accuracy to 7.0% and inducing class-level confusions, such as misclassifying giraffes and elephants as houses. On the other hand, Task 3’s DAN-DG aligned the three source domains without target data and led to a better target accuracy (+13.92%). However, a high value of $\lambda \geq 1$ for Task 3 led to a decrease in Sketch accuracy, confirming that excessive alignment can lead to class-level mixing and, ultimately, lower target performance.

Unlabeled target access caused negative transfer due to the uneven class distribution of the target images (e.g., a -10.4% accuracy drop in DAN, compared to target-free DAN-DG’s +13.92% gain), which forced target images into the wrong class boundaries that were more abundant in the source. However, this is not sufficient to definitively conclude that target access is strictly worse than

withholding target data. Our results were highly influenced by the specific dataset bias, baseline accuracy differences between Task 2 and Task 3, and the fact that DAN-DG computed its Maximum Mean Discrepancy (MMD) by averaging across three distinct source domains.

4.3 Recognition versus Rejection in Open Sets

Task 4 demonstrates that representations relying solely on ordinary classification losses do not necessarily become safer under semantic novelty. Driven merely by classification accuracy, models tend to learn the easiest available shortcuts, developing spurious correlations (e.g., predicting a cat based on a blue background rather than the animal’s features). This phenomenon is evident in the baseline setup: even though the Vanilla model achieves a 94.61% closed-set classification accuracy (CSA), its reliance on shortcuts led to confident misclassifications on far unknowns (e.g., confusing a bottle for a bird with a highly confident MLS score of -8.05).

GCSC demonstrates that improving dataset informativeness via strong random augmentations can remove some of these spurious correlations, forcing the model to learn causal features. This improved both CSA (95.33%) and far-unknown rejection (AUROC jumped from 0.883 to 0.915). However, it is still insufficient to completely eliminate near-end confusion, as near-unknown features naturally resemble actual class features. Conversely, PROSER showed that explicitly assuming an inherent structure for the target features relative to the source can lead to worse results, degrading both known accuracy (CSA dropped to 94.42%) and unknown rejection (near AUROC fell to 0.786), likely because its dummy classifiers overlapped excessively with true class boundaries. Ultimately, confusing near unknowns is an expected limitation of semantic proximity, whereas confusing far unknowns stems from spurious correlations that arise when a dataset is not informative enough to prevent shortcut learning.

4.4 Overall Synthesis

Across all four tasks, it is evident that a robust visual representation should preserve global shape rather than relying on local texture. Task 1 illustrated that architectures penalized for misalignment (e.g., CLIP) inherently learn global structures, whereas standard models like ResNet default to learning local shortcuts to achieve correct classification. Tasks 2 and 3 revealed the danger of forcing domain invariance without considering underlying distributions. Both CDAN and DAN induced severe negative transfer because they attempted to match feature distributions while ignoring the mismatched class priors between the source and target domains. Task 3 showed that averaging alignment across multiple distinct source domains can lead to better generalization. Finally, Task 4 established that even when a model successfully learns invariant features, inputs that fall far outside the semantic distribution of the training data must be treated as outside the label space. A robust architecture therefore requires a combination of heavy augmentation to learn causal features, domain generalization to handle stylistic shifts, and distance-based thresholding (like Mahalanobis) to safely reject what it has never seen.

5 Conclusion

In conclusion, this report demonstrates that a robust visual representation must capture global shape and structure rather than relying on local textures or spurious shortcuts. While text-supervised architectures naturally develop better structural representations, standard classification models often take the easy route of memorizing simple visual cues. Furthermore, blindly forcing domain invariance without accounting for the underlying class distributions can cause severe negative transfer. Averaging alignment across multiple source domains and utilizing heavy data augmentations prove to be far more effective strategies. Ultimately, true robustness requires models that can both generalize across stylistic shifts and safely reject inputs that fall entirely outside their learned label space.

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A Appendix

A.1 Task 2: Supplementary Material

Since the mean values can give a wrong idea about the source accuracies, Table 5 (Section 3.2) shows that all models achieve a similar accuracy on all domains, with art painting having the least accuracy. This can also serve as a reason for our lower target accuracy, as art paintings can be thought of as somewhat more similar to sketches relative to photos and cartoons.

Figure 8 displays the class-wise distribution for both the source and target. This shows us why even CDAN fails in having a higher target accuracy. Figure 9 indicates class separability and domain separability across all domains. It can be seen that apart from DAN, none of the models achieve any significant domain mixing, whereas CDAN mixes the classes of our test as well.

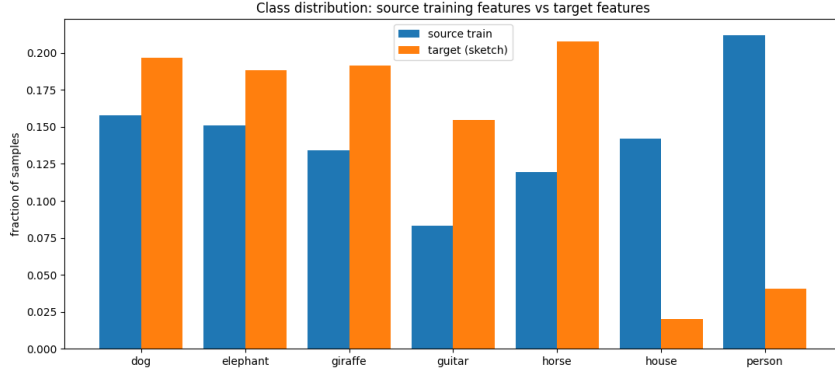


Figure 8: Class distribution comparison between the pooled source training features and the target (sketch) features.

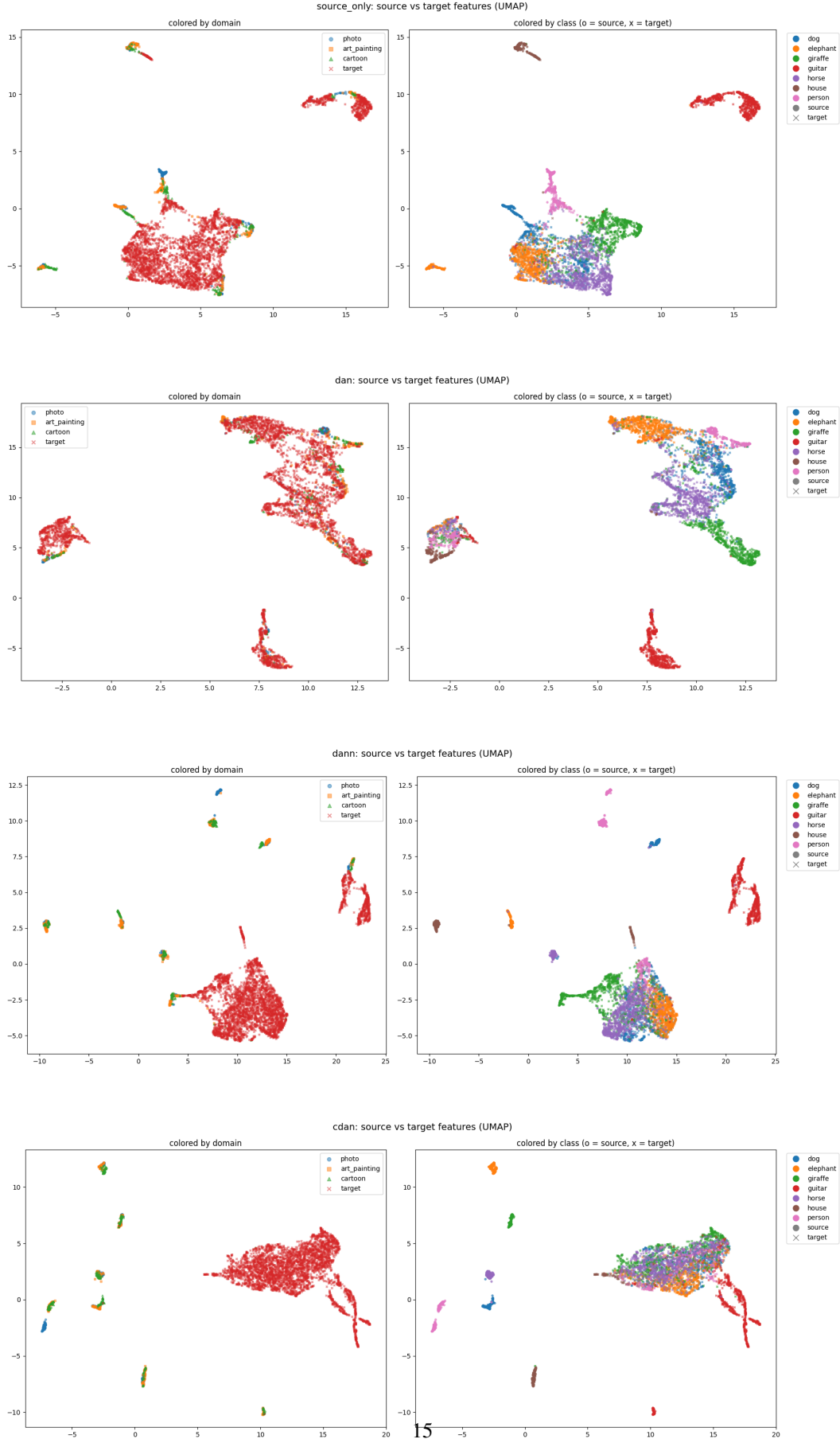


Figure 9: UMAP projections of source and target features in the latent space. **Top to bottom:** Source-only, DAN, DANN, and CDAN.